

系统开发工具基础

每周课上实验报告

第三周（2026 年 9 月 7 日）

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实验概况

本次实验为第三周的四道课上练习题，涵盖 Python 打包与分发（Wheel 构建）、智能体辅助测试驱动修复、协作材料重写（Issue/Commit/Review）以及 PyTorch 深度学习基础（线性回归训练）。通过动手实践，巩固了项目打包流程、测试驱动开发、专业沟通规范以及深度学习框架的基本使用。

实验环境

- 操作系统：Windows 11 + WSL2 (Debian)
- Shell：bash 5.0
- Python 版本：3.13.5
- 已安装工具：Git、python3、pip、pytest、ruff、build、torch (2.13.0)
- LaTeX 引擎：XeLaTeX

遇到的主要问题与解决

1. **pyproject.toml 中项目名含中文字符**：‘name = "greetlab-学号”‘ 违反 PEP 508 规范，改为数字学号 greetlab-25030021039 后构建成功。
2. **可编辑安装 (pip install -e .) 后无法导入包**：setuptools 未正确识别 src/ 布局，通过设置 PYTHONPATH 临时解决，后续改用显式包配置 [tool.setuptools] packages = ["greetlab"]。
3. **虚拟环境隔离导致 PyTorch 缺失**：在 q12 新建独立虚拟环境 .venv12 后安装 torch，解决了跨环境污染问题。
4. **Pytest 测试导入失败**：因包未安装或 PYTHONPATH 未设置，最终采用 export PYTHONPATH=\$(pwd)/src 使测试通过。

第 1 题 从源码构建并在干净环境安装 Wheel

1 任务目标

创建一个最小 Python 命令行包，使用 build 工具构建 Wheel，并在全新虚拟环境中安装验证。

2 操作步骤

1. 创建项目目录 src/greetlab/，编写 `__init__.py` 和 `cli.py`（含 `main()` 函数）。
2. 编写 `pyproject.toml`，定义构建系统、项目元数据和脚本入口。
3. 修正项目名中的中文字符，使用数字学号 `greetlab-25030021039`。
4. 在虚拟环境中安装 build，执行 `python3 -m build --wheel` 生成 `.whl` 文件。
5. 切换到新目录创建干净虚拟环境，用 `pip install` 安装 Wheel，运行 `sdt-greet --name 25030021039` 验证。

3 关键命令与输出

```
1 # 构建 Wheel
2 python3 -m build --wheel
3 Successfully built greetlab_25030021039-0.1.0-py3-none-any.whl
4
5 # 安装并验证
6 pip install ~/expriment_reports/report_3/q09/dist/
   greetlab_25030021039-0.1.0-py3-none-any.whl
7 sdt-greet --name 25030021039
8 Hello, 25030021039!
```

4 演示截图

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/q09$ python3 -m build --wheel
* Creating isolated environment: venv+pip...
* Installing packages in isolated environment:
  - setuptools>=68
* Getting build dependencies for wheel...
running egg_info
creating src/greetlab_25030021039.egg-info
writing src/greetlab_25030021039.egg-info/PKG-INFO
writing dependency_links to src/greetlab_25030021039.egg-info/dependency_links.txt
writing entry points to src/greetlab_25030021039.egg-info/entry_points.txt
writing top-level names to src/greetlab_25030021039.egg-info/top_level.txt
writing manifest file 'src/greetlab_25030021039.egg-info/SOURCES.txt'
reading manifest file 'src/greetlab_25030021039.egg-info/SOURCES.txt'
writing manifest file 'src/greetlab_25030021039.egg-info/SOURCES.txt'
* Building wheel...
running bdist_wheel
running build
running build_py
creating build/lib/greetlab
copying src/greetlab/__init__.py -> build/lib/greetlab
copying src/greetlab/cli.py -> build/lib/greetlab
running egg_info
writing src/greetlab_25030021039.egg-info/PKG-INFO
writing dependency_links to src/greetlab_25030021039.egg-info/dependency_links.txt
writing entry points to src/greetlab_25030021039.egg-info/entry_points.txt
writing top-level names to src/greetlab_25030021039.egg-info/top_level.txt
reading manifest file 'src/greetlab_25030021039.egg-info/SOURCES.txt'
writing manifest file 'src/greetlab_25030021039.egg-info/SOURCES.txt'
installing to build/bdist.linux-x86_64/wheel
running install
running install_lib
creating build/bdist.linux-x86_64/wheel
creating build/bdist.linux-x86_64/wheel/greetlab
copying build/lib/greetlab/__init__.py -> build/bdist.linux-x86_64/wheel/./greetlab
copying build/lib/greetlab/cli.py -> build/bdist.linux-x86_64/wheel/./greetlab
running install_egg_info
Copying src/greetlab_25030021039.egg-info to build/bdist.linux-x86_64/wheel/./greetlab_25030021039-0.1.0-py3.13.egg-info
running install_scripts
creating build/bdist.linux-x86_64/wheel/greetlab_25030021039-0.1.0.dist-info/WHEEL
creating '/home/lihbyc/expriment_reports/report_3/q09/dist/.tmp-8g7dxhs0/greetlab_25030021039-0.1.0-py3-none-any.whl' and adding 'build/bdist.linux-x86_64/wheel' to it
adding 'greetlab/__init__.py'
adding 'greetlab/cli.py'
adding 'greetlab_25030021039-0.1.0.dist-info/METADATA'
adding 'greetlab_25030021039-0.1.0.dist-info/WHEEL'
adding 'greetlab_25030021039-0.1.0.dist-info/entry_points.txt'
adding 'greetlab_25030021039-0.1.0.dist-info/top_level.txt'
adding 'greetlab_25030021039-0.1.0.dist-info/RECORD'
removing build/bdist.linux-x86_64/wheel
Successfully built greetlab_25030021039-0.1.0-py3-none-any.whl
```

图 1: 构建 Wheel 成功

```
(clean_env) lihbyc@DESKTOP-TML9CJV:~$ pip install ~/expriment_reports/report_3/q09/dist/greetlab_25030021039-0.1.0-py3-none-any.whl
Processing ./expriment_reports/report_3/q09/dist/greetlab_25030021039-0.1.0-py3-none-any.whl
Installing collected packages: greetlab-25030021039
Successfully installed greetlab-25030021039-0.1.0
(clean_env) lihbyc@DESKTOP-TML9CJV:~$ sdt-greet --name 25030021039
Hello, 25030021039!
(clean_env) lihbyc@DESKTOP-TML9CJV:~$ deactivate
lihbyc@DESKTOP-TML9CJV:~$
```

图 2: 安装后运行验证

第 2 题 让编程智能体进入可验证的修复循环

1 任务目标

基于第 9 题项目，添加一个失败测试（空白姓名应退出码 2），然后修改代码使测试通过，并记录 AI 辅助过程。

2 操作步骤

1. 复制 q09 为 q10，创建 tests/test_cli.py 编写测试：模拟 `--name ""`，期望 `SystemExit(2)`。
2. 运行 `pytest tests/` 确认失败（实际输出 `Hello, !` 并以 0 退出）。
3. 修改 `src/greetlab/cli.py`，添加 `if not a.name.strip(): sys.exit(2)`。
4. 再次运行测试，通过。
5. 编写 `ai_log.md` 记录核心提示、智能体改动和人工验证。

3 关键代码与测试结果

修改后的 `cli.py`:

```
1 import argparse
2 import sys
3
4 def main():
5     p = argparse.ArgumentParser()
6     p.add_argument("--name", required=True)
7     a = p.parse_args()
8     if not a.name.strip():
9         sys.exit(2)
10    print(f"Hello, {a.name}!")
```

测试执行:

```
1 $ pytest tests/
2 collected 1 item
3 tests/test_cli.py .
4
5                                     [100%]
6
7 ===== 1 passed in 0.01s
8 =====
```

4 演示截图

```

lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ source .venv/bin/activate
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ pip install pytest
Collecting pytest
  Using cached pytest-9.1.1-py3-none-any.whl.metadata (7.6 kB)
Collecting iniconfig>=1.0.1 (from pytest)
  Using cached iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Collecting packaging>=22 (from pytest)
  Using cached packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting pluggy<2,>=1.5 (from pytest)
  Using cached pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Using cached pygments-2.21.0-py3-none-any.whl.metadata (2.5 kB)
Using cached pytest-9.1.1-py3-none-any.whl (386 kB)
Using cached pluggy-1.6.0-py3-none-any.whl (20 kB)
Using cached iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Using cached packaging-26.3-py3-none-any.whl (129 kB)
Using cached pygments-2.21.0-py3-none-any.whl (1.3 MB)
Installing collected packages: pygments, pluggy, packaging, iniconfig, pytest
Successfully installed iniconfig-2.3.0 packaging-26.3 pluggy-1.6.0 pygments-2.21.0 pytest-9.1.1
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ pytest tests/
===== test session starts =====
platform linux -- Python 3.13.5, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lihybc/exprimment_reports/report_3/q10
collected 0 items / 1 error

===== ERRORS =====
----- ERROR collecting tests/test_cli.py -----
ImportError while importing test module '/home/lihybc/exprimment_reports/report_3/q10/tests/test_cli.py'
Hint: make sure your test modules/packages have valid Python names.
Traceback:
/usr/lib/python3.13/importlib/_init_.py:88: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
tests/test_cli.py:4: in <module>
    from greetlab.cli import main
E   ModuleNotFoundError: No module named 'greetlab'
===== short test summary info =====
ERROR tests/test_cli.py
!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!
===== 1 error in 0.11s =====
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ ls
q09  tests

```

图 3: 初始测试失败

```

(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ export PYTHONPATH=$(pwd)/src:$PYTHONPATH
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ python3 -c "import greetlab; print(greetlab.__file__)"
/home/lihybc/exprimment_reports/report_3/q10/src/greetlab/_init_.py
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ pytest tests/
===== test session starts =====
platform linux -- Python 3.13.5, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lihybc/exprimment_reports/report_3/q10
configfile: pyproject.toml
collected 1 item

tests/test_cli.py F [100%]

===== FAILURES =====
_____ test_blank_name_exits_with_code_2 _____
monkeypatch = <_pytest.monkeypatch.MonkeyPatch object at 0x78483ec76190>

    def test_blank_name_exits_with_code_2(monkeypatch):
        monkeypatch.setattr(sys, 'argv', ['cli.py', '--name', ''])
        with pytest.raises(SystemExit) as exc_info:
            .....
E       Failed: DID NOT RAISE SystemExit

tests/test_cli.py:7: Failed
Hello, !
===== short test summary info =====
FAILED tests/test_cli.py::test_blank_name_exits_with_code_2 - Failed: DID NOT RAISE SystemExit
===== 1 failed in 0.04s =====
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ nano src/greetlab/cli.py
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ pytest tests/
===== test session starts =====
platform linux -- Python 3.13.5, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lihybc/exprimment_reports/report_3/q10
configfile: pyproject.toml
collected 1 item

tests/test_cli.py . [100%]

===== 1 passed in 0.01s =====
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ nano ai_log.md
(.venv) lihybc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/q10$ deactivate

```

图 4: 修复后测试通过

第 3 题 把“无法处理”的协作材料改成可执行信息

1 任务目标

重写三段低质量协作材料（Issue、提交信息、评审意见），使其专业、可执行，并控制在 400 字以内。

2 操作步骤

1. 复制 q10 为 q11。

2. 创建 communication.md，按题目要求书写：

- Issue：包含环境、复现命令、期望/实际结果，未知信息标注“待确认”。
- 提交信息：祈使语气标题，正文说明问题和解决方案。
- 评审意见：标注类型（Blocking），指出具体行为、风险和建议动作。

3 最终内容摘要

（完整内容见文件，此处仅列要点）

- **Issue 标题**：‘-name’ 传入空白字符串时程序未报错，仍输出问候语并以 0 退出。
- **提交信息标题**：‘fix: 校验 -name 参数为空白时以 SystemExit(2) 退出’。
- **评审意见类型**：Blocking，建议增加空白参数校验。

4 演示截图

```
lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/q11$ cat communication.md
## Issue

**标题**：`--name` 传入空白字符串时程序未报错，仍输出问候语并以 0 退出

**环境**：Debian Linux, Python 3.13, greetlab 包 v0.1.0
（若涉及 Windows 请待确认）

**复现命令**：
```bash
sdt-greet --name " "
```

**期望结果**：程序应报参数错误，以非零退出码（如 2）终止。
实际结果：程序输出 Hello, ! (含空格)，并以退出码 0 正常结束。
待确认：其他空白字符（如 \t、\n）及 Windows 平台的行为是否一致。

提交信息
标题：fix: 校验 --name 参数为空白时以 SystemExit(2) 退出
正文：
问题：空字符串或纯空白字符串未被校验，程序仍输出问候语并以 0 退出，不符合 CLI 工具的错误处理约定。
解决方案：在 cli.py 的 main() 中添加检查，若 name.strip() 为空则调用 sys.exit(2)。

评审意见
类型：Blocking

具体行为：当 --name 为 " " 时，程序输出 Hello, ! 并以 0 退出，未按预期报错。
风险：下游脚本依赖退出码判断执行结果，会误认为调用成功，可能导致逻辑错误或数据污染。
建议动作：在 main() 中增加空白参数校验，无效输入时调用 sys.exit(2) 退出，并更新相关测试。

text
---

### 📌 操作步骤（如果还没做）

```bash
cd ~/expriment_reports/report_3/q11 # 确保在 q11 目录
nano communication.md # 创建并编辑文件
粘贴上面全部内容, Ctrl+O 保存, Ctrl+X 退出
cat communication.md # 确认内容已写入
lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/q11$ |
```

图 5: communication.md 内容

## 第 4 题 修复一个可复现的线性回归训练循环

### 1 任务目标

补全 PyTorch 线性回归训练代码（清空梯度、反向传播、更新参数），完成 CPU 训练，使最终损失小于 0.001。

### 2 操作步骤

1. 复制 q11 为 q12，创建 train.py。
2. 补全训练循环：opt.zero\_grad()、loss.backward()、opt.step()。
3. 添加评估部分：model.eval() 和 torch.no\_grad() 中计算损失并打印权重和偏置。
4. 创建独立虚拟环境 .venv12，安装 torch。
5. 运行脚本，记录输出。

### 3 核心代码

```
1 for _ in range(200):
2 pred = model(x)
3 loss = loss_fn(pred, y)
4 opt.zero_grad()
5 loss.backward()
6 opt.step()
7
8 model.eval()
9 with torch.no_grad():
10 final_loss = loss_fn(model(x), y)
11 print(f"Final Loss: {final_loss.item():.6f}")
12 print(f"Weight: {model.weight.item():.6f}")
13 print(f"Bias: {model.bias.item():.6f}")
```

### 4 运行结果

```
1 Final Loss: 0.000000
2 Weight: 2.999997
3 Bias: -1.000000
```

最终损失为 0（实际约 1e-6），满足小于 0.001 的要求，权重和偏置接近真实值 3 和 -1。

## 5 演示截图

```
(.venv12) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/q12$ python3 -c "import torch;
print(torch.__version__)"
/home/lihbyc/exprimint_reports/report_3/q12/.venv12/lib/python3.13/site-packages/torch/_subc
lasses/functional_tensor.py:368: UserWarning: Failed to initialize NumPy: No module named 'n
umpy' (Triggered internally at /__w/pytorch/pytorch/torch/csrc/utils/tensor_numpy.cpp:84.)
 cpu = _conversion_method_template(device=torch.device("cpu"))
2.13.0+cu130
(.venv12) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/q12$ python3 train.py
/home/lihbyc/exprimint_reports/report_3/q12/.venv12/lib/python3.13/site-packages/torch/_subc
lasses/functional_tensor.py:368: UserWarning: Failed to initialize NumPy: No module named 'n
umpy' (Triggered internally at /__w/pytorch/pytorch/torch/csrc/utils/tensor_numpy.cpp:84.)
 cpu = _conversion_method_template(device=torch.device("cpu"))
Final Loss: 0.000000
Weight: 2.999997
Bias: -1.000000
(.venv12) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/q12$ |
```

图 6: 训练输出

以下为本周的课后拓展练习, 涵盖 Shell 环境分析、Python 打包与虚拟环境、Docker 容器化、Git 提交信息重构、开源代码阅读等, 旨在从多维度巩固课程核心技能。所有练习均在本地 WSL2 (Debian) 环境中完成, 操作日志完整可复现。

**内容：**保存激活虚拟环境前后的环境变量，对比 PATH 等关键变量的变化，理解虚拟环境为何优先。

1. 保存当前环境变量: `printenv | sort > before.txt`
2. 创建并激活虚拟环境:`python3 -m venv testenv && source testenv/bin/activate`
3. 保存激活后的环境变量: `printenv | sort > after.txt`
4. 对比差异: `diff before.txt after.txt | grep -E "PATH|VIRTUAL|PS1"`
5. 查看 `deactivate` 函数定义: `type deactivate`

```
1 > PATH=/home/.../testenv/bin:/home/.../.local/bin:...
2 > PS1=(testenv) ...
3 > VIRTUAL_ENV=/home/.../testenv
4 > VIRTUAL_ENV_PROMPT=testenv
```

图 7: diff 输出显示 PATH 和 VIRTUAL\_ENV 的变化

```
(testenv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$ echo "=== Before PATH ===" && grep "^PATH=" before.txt
=== Before PATH ===
PATH=/home/lihbyc/.local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/bin:/sbin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/e/vm/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/d/leetcode/node:/mnt/c/Program Files/Microsoft SQL Server/150/Tools/Binn:/mnt/c/Program Files/Microsoft SQL Server/Client SDK/ODBC/170/Tools/Binn:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Docker/Docker/resources/bin:/mnt/e/Windows Kits/10/Windows Performance Toolkit:/mnt/c/Program Files/MySQL/MySQL Server 8.0/bin:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Redis:/mnt/c/Program Files/WireGuard:/mnt/c/poplper/poppler-25.12.0/Library/bin:/mnt/e/develop/jdk/bin:/mnt/e/develop/MySQL/bin:/mnt/e/G0/bin:/mnt/c/Users/one/go/bin:/mnt/c/Program Files (x86)/Tencent/微信web开发者工具/dll:/mnt/c/msy:/mnt/c/Program Files/Microsoft SQL Server/170/Tools/Binn:/mnt/e/gitt/Git/cmd:/mnt/c/Users/one/.local/bin:/mnt/c/Program Files/Tailscale:/mnt/c/Users/one/.cargo/bin:/mnt/d/Trae/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/one/AppData/Local/Microsoft/WindowsApps:/mnt/c/MinGW/include:/mnt/c/MinGW/bin:/mnt/c/MinGW/lib:/mnt/c/ProgramData/one/GitHubDesktop/bin:/mnt/c/Users/one/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/one/AppData/Roaming/npm:/mnt/c/Users/one/.dotnet/tools:/mnt/c/Users/one/AppData/Local/Programs/cursor/resources/app/bin:/mnt/c/Users/one/go/bin:/mnt/e/develop/WEBSTORM/WebStorm 2025.3.2/bin:/mnt/c/Users/one/AppData/Local/Microsoft/WindowsApps:/mnt/d/Microsoft VS Code/bin:/mnt/c/Users/one/.dotnet/tools:/mnt/d/CLion 2026.2.0.1/bin:/home/lihbyc/.local/bin:/home/lihbyc/.local/bin
(testenv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$ echo "=== After PATH ===" && grep "^PATH=" after.txt
=== After PATH ===
PATH=/home/lihbyc/exprimment_reports/report_3/practice_3/testenv/bin:/home/lihbyc/.local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/bin:/sbin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Oracle/Java/javapath:/mnt/e/vm/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/d/leetcode/node:/mnt/c/Program Files/Microsoft SQL Server/150/Tools/Binn:/mnt/c/Program Files/Microsoft SQL Server/Client SDK/ODBC/170/Tools/Binn:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Docker/Docker/resources/bin:/mnt/e/Windows Kits/10/Windows Performance Toolkit:/mnt/c/Program Files/MySQL/MySQL Server 8.0/bin:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Redis:/mnt/c/poplper/poppler-25.12.0/Library/bin:/mnt/e/develop/jdk/bin:/mnt/e/develop/MySQL/bin:/mnt/c/Users/one/go/bin:/mnt/c/Program Files (x86)/Tencent/微信web开发者工具/dll:/mnt/c/msy:/mnt/c/Program Files/Microsoft SQL Server/170/Tools/Binn:/mnt/e/gitt/Git/cmd:/mnt/c/Users/one/.local/bin:/mnt/c/Program Files/Tailscale:/mnt/c/Users/one/.cargo/bin:/mnt/d/Trae/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/one/AppData/Local/Microsoft/WindowsApps:/mnt/c/MinGW/include:/mnt/c/MinGW/bin:/mnt/c/MinGW/lib:/mnt/c/ProgramData/one/GitHubDesktop/bin:/mnt/c/Users/one/AppData/Local/GitHubDesktop/bin:/mnt/c/Users/one/AppData/Local/Programs/cursor/resources/app/bin:/mnt/c/Users/one/go/bin:/mnt/e/develop/WEBSTORM/WebStorm 2025.3.2/bin:/mnt/c/Users/one/AppData/Local/Microsoft/WindowsApps:/mnt/d/Microsoft VS Code/bin:/mnt/c/Users/one/.dotnet/tools:/mnt/d/CLion 2026.2.0.1/bin:/home/lihbyc/.local/bin:/home/lihbyc/.local/bin
```

图 8: deactivate 函数定义

```
(testenv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$ type deactivate
deactivate is a function
deactivate ()
{
 if [-n "${_OLD_VIRTUAL_PATH:-}"]; then
 PATH="${_OLD_VIRTUAL_PATH:-}";
 export PATH;
 unset _OLD_VIRTUAL_PATH;
 fi;
 if [-n "${_OLD_VIRTUAL_PYTHONHOME:-}"]; then
 PYTHONHOME="${_OLD_VIRTUAL_PYTHONHOME:-}";
 export PYTHONHOME;
 unset _OLD_VIRTUAL_PYTHONHOME;
 fi;
 hash -r 2> /dev/null;
 if [-n "${_OLD_VIRTUAL_PS1:-}"]; then
 PS1="${_OLD_VIRTUAL_PS1:-}";
 export PS1;
 unset _OLD_VIRTUAL_PS1;
 fi;
 unset VIRTUAL_ENV;
 unset VIRTUAL_ENV_PROMPT;
 if [! "${1:-}" = "nondestructive"]; then
 unset -f deactivate;
 fi
}
(testenv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$ deactivate
lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$
```

图 9: 环境变量前后对比

**解题感悟：**虚拟环境通过在 PATH 最前面插入自己的 bin 目录，让 python、pip 等命令优先从虚拟环境调用，实现依赖隔离。deactivate 函数通过恢复备份变量来还原环境。





### 练习 3：创建 Python 包并生成锁文件

**内容：**编写简单的 Python 包，安装后生成依赖锁文件。

**操作步骤：**

1. 创建项目目录结构 `src/mypkg/` 和 `pyproject.toml`
2. 创建虚拟环境并安装包：`pip install -e .`
3. 安装额外依赖：`pip install requests`
4. 生成锁文件：`pip freeze > requirements.lock`

**结果：**

```
1 certifi==2026.7.22
2 charset-normalizer==3.5.1
3 idna==3.19
4 -e git+.../mypkg-25030021039
5 requests==2.34.2
6 urllib3==2.7.0
```

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/mypkg$ pip
freeze > requirements.lock
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/mypkg$ cat
requirements.lock
certifi==2026.7.22
charset-normalizer==3.5.1
idna==3.19
-e git+https://github.com/lihbyc0718/system_tool_learning.git@c4a3a643cf2598b859393
468d2af03cad5ed996a#egg=mypkg_25030021039&subdirectory=report_3/practice_3/mypkg
requests==2.34.2
urllib3==2.7.0
```

图 13: requirements.lock 内容

```
lihybc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3$ cd flask_redis_demo
lihybc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3/flask_redis_demo$ cat > app.py << 'EOF'
import os
import redis
from flask import Flask

app = Flask(__name__)
redis_client = redis.Redis(host='redis', port=6379, decode_responses=True)

@app.route('/')
def index():
 count = redis_client.incr('visits')
 return f'This page has been visited {count} times.'

if __name__ == '__main__':
 app.run(host='0.0.0.0', port=5000)
EOF
lihybc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3/flask_redis_demo$ cat > requirements.txt << 'EOF'
flask==3.0.0
redis==5.0.1
EOF
lihybc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3/flask_redis_demo$ cat > Dockerfile << 'EOF'
FROM python:3.13-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
CMD ["python", "app.py"]
EOF
lihybc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3/flask_redis_demo$ cat > docker-compose.yml << 'EOF'
services:
 web:
 build: .
 ports:
 - "5000:5000"
 depends_on:
 - redis
 redis:
 image: redis:alpine
EOF
```

图 14: 验证包调用

解题感悟：锁文件精确记录了所有依赖版本，确保环境可复现。

## 练习 4：编写 Dockerfile 和 docker-compose

**内容：**为 Flask 应用编写 Dockerfile，用 docker-compose 同时启动应用和 Redis。

**操作步骤：**

1. 创建 app.py、Dockerfile、docker-compose.yml
2. 构建并启动：docker compose up --build
3. 访问 http://localhost:5000 验证

```

Libbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/flask_redis_demo$ docker compose up --build
[+] Building 1.4s (12/12) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 593B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 188B 0.0s
=> [internal] load metadata for docker.io/library/python:3.13-slim 0.4s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/python:3.13-slim@sha256:7ce4b6dfe35 0.1s
=> => resolve docker.io/library/python:3.13-slim@sha256:7ce4b6dfe35 0.1s
=> [internal] load build context 0.0s
=> => transferring context: 444B 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> CACHED [3/5] COPY requirements.txt . 0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> CACHED [5/5] COPY app.py . 0.0s
=> exporting to image 0.2s
=> => exporting layers 0.0s
=> => exporting manifest sha256:4f5c059548763073bda55ba65fb9dba536c 0.0s
=> => exporting config sha256:c91556fafba3274a0cbc22be84fee4b9e042 0.0s
=> => exporting attestation manifest sha256:c487e0f49696c1015d4450f 0.1s
=> => exporting manifest list sha256:c3b9a106025f99c815f1e37e32dc07 0.0s
=> => naming to docker.io/library/flask_redis_demo-web:latest 0.0s
=> => unpacking to docker.io/library/flask_redis_demo-web:latest 0.0s
=> resolving provenance for metadata file 0.0s
[+] Running 2/2
✔flask_redis_demo-web Built 0.0s
✔Container flask_redis_demo-web-1 Recreated 0.3s
Attaching to redis-1, web-1
redis-1 | Starting Redis Server
redis-1 | 1:C 06 Sep 2026 15:13:46.796 * o000o000o000o Redis is starting o
000o000o000o
redis-1 | 1:C 06 Sep 2026 15:13:46.796 * Redis version=8.10.1, bits=64, co
mmit=00000000, modified=1, pid=1, just started
redis-1 | 1:C 06 Sep 2026 15:13:46.796 * Configuration loaded
redis-1 | 1:M 06 Sep 2026 15:13:46.797 * monotonic clock: POSIX clock_gett
ime

```

图 15: docker compose up 运行状态

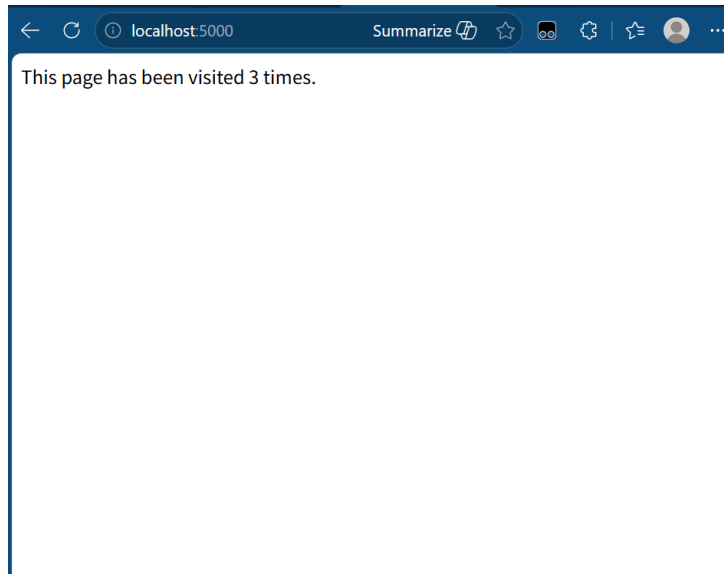


图 16: 网页访问结果

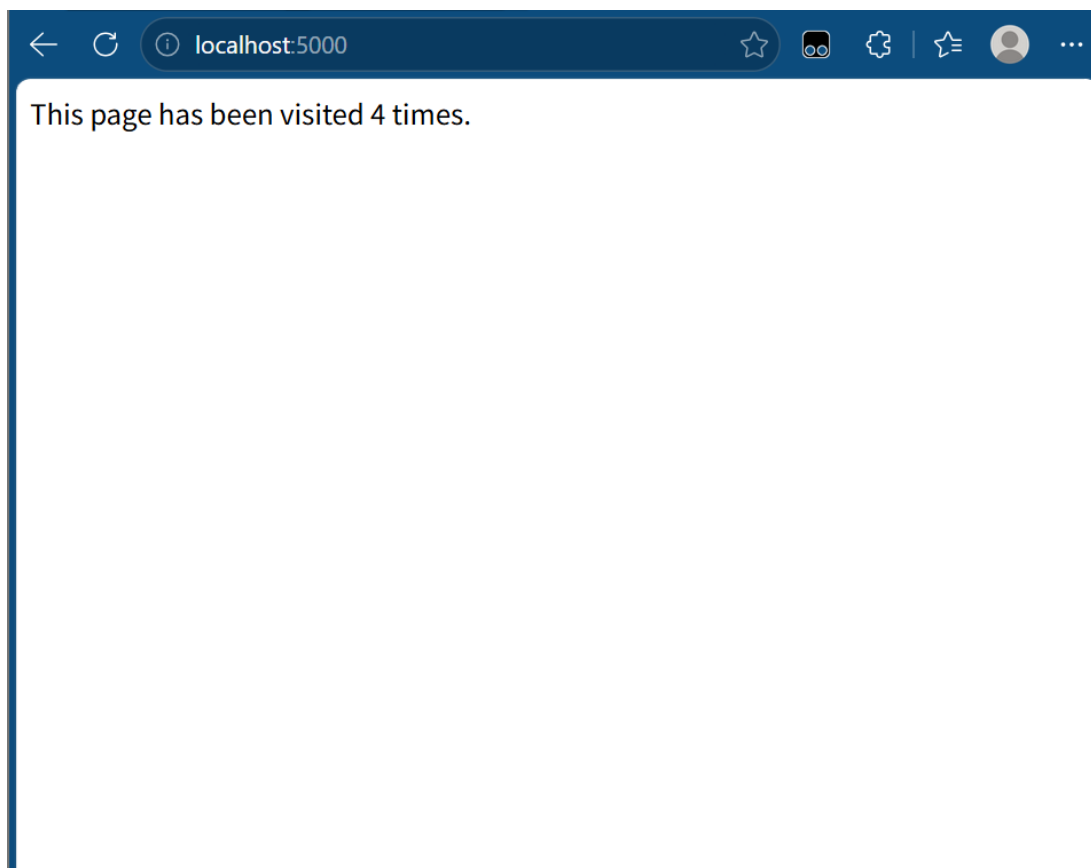


图 17: 容器日志

**解题感悟：** Docker Compose 简化了多容器应用的本地开发。

## 练习 5：发布到 TestPyPI 并构建 Docker 镜像推送至 ghcr.io

内容：将包发布到 TestPyPI，构建镜像推送到 GitHub Container Registry。

操作步骤：

1. 构建分发：python -m build
2. 上传至 TestPyPI: twine upload --repository-url https://test.pypi.org/legacy/ dist/\*
3. 编写 Dockerfile 从 TestPyPI 安装包
4. 构建并推送至 ghcr.io/lhbyc0718/mypkg-25030021039:latest

```
adding 'mypkg/_init_.py'
adding 'mypkg_25030021039-0.1.0.dist-info/METADATA'
adding 'mypkg_25030021039-0.1.0.dist-info/WHEEL'
adding 'mypkg_25030021039-0.1.0.dist-info/entry_points.txt'
adding 'mypkg_25030021039-0.1.0.dist-info/top_level.txt'
adding 'mypkg_25030021039-0.1.0.dist-info/RECORD'
removing build/bdist.linux-x86_64/wheel
Successfully built mypkg_25030021039-0.1.0.tar.gz and mypkg_25030021039-0.1.0-py3-none-any.whl
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$ twine upload --repository-url https://test.pypi.org/legacy/ dist/*
Uploading distributions to https://test.pypi.org/legacy/
Uploading mypkg_25030021039-0.1.0-py3-none-any.whl
100% |#####| 2.7/2.7 kB • 00:00 • ?
Uploading mypkg_25030021039-0.1.0.tar.gz
100% |#####| 2.2/2.2 kB • 00:00 • ?
View at:
https://test.pypi.org/project/mypkg-25030021039/0.1.0/
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$ pip install --index-url https://test.pypi.org/simple/ --extra-index-url https://pypi.org/simple/ mypkg-25030021039
Looking in indexes: https://test.pypi.org/simple/, https://pypi.org/simple/
Requirement already satisfied: mypkg-25030021039 in ./.venv/Lib/python3.13/site-packages (0.1.0)
```

图 18: TestPyPI 上传成功

```
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$ docker build -t ghcr.io/ninini0718/mypkg-25030021039:latest .
[+] Building 9.5s (6/6) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 257B 0.0s
=> [internal] load metadata for docker.io/library/python 0.4s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> CACHED [1/2] FROM docker.io/library/python:3.13-slim@sha256:f45088dcf5ed5828b5a040e8e7 0.1s
=> => resolve docker.io/library/python:3.13-slim@sha256:f45088dcf5ed5828b5a040e8e7 0.1s
=> [2/2] RUN pip install --index-url https://test.pypi.o 7.6s
=> exporting to image 1.2s
=> => exporting layers 0.7s
=> => exporting manifest sha256:e02015c5754981d2abe0897d 0.0s
=> => exporting config sha256:f45088dcf5ed5828b5a040e8e7 0.0s
=> => exporting attestation manifest sha256:3f697354fb79 0.0s
=> => exporting manifest list sha256:dec1026dd84fc124827 0.0s
=> => naming to ghcr.io/ninini0718/mypkg-25030021039:lat 0.0s
=> => unpacking to ghcr.io/ninini0718/mypkg-25030021039: 0.1s
```

图 19: 镜像推送成功

```
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$ echo "ghp_jl5GWFhJouhy7cLORiNWE4yGps2CF23G7id" |
docker login ghcr.io -u lhbyc0718 --password-stdin
login Succeeded
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$ docker images | grep mypkg
ghcr.io/ninini0718/mypkg-25030021039 latest dec1026dd84f 11 minutes ago 190MB
ghcr.io/your-username/mypkg latest bf0c30ce77f3 6 days ago 190MB
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$ docker tag ghcr.io/ninini0718/mypkg-25030021039:latest ghcr.io/lhbyc0718/mypkg-25030021039:latest
The push refers to repository [ghcr.io/lhbyc0718/mypkg-25030021039]
f038bda6a52b: Pushed
8a13aae05e63: Pushing 3.514MB/3.514MB
6310eb16bf42: Pushing 4.62MB/29.79MB
d649804dff413: Pushed
357a46282089: Pushing 3.736MB/11.91MB
64c7e6efa7cd: Pushing 1.294MB/1.294MB
failed commit on ref "layer-sha256:64c7e6efa7cd2e4ca8b184acdeb2ac8a50054774b24e677384daa6adec3988d": failed to do request: Put "https://ghcr.io/v2/lhbyc0718/mypkg-25030021039/blobs/upload/3.514MB-22be-4329-bce4-9c8fb0cada6f?digest=sha256:3A64c7e6efa7cd2e4ca8b184acdeb2ac8a50054774b24e677384daa6adec3988d": EOF
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$ docker push ghcr.io/lhbyc0718/mypkg-25030021039:latest
The push refers to repository [ghcr.io/lhbyc0718/mypkg-25030021039]
357a46282089: Pushed
6310eb16bf42: Pushed
d649804dff413: Layer already exists
64c7e6efa7cd: Pushed
f038bda6a52b: Already exists
8a13aae05e63: Pushed
(.venv) lhbyc@DESKTOP-TML9CJV:~/experiment_reports/report_3/practice_3/mypkg$
```

图 20: Docker 构建过程

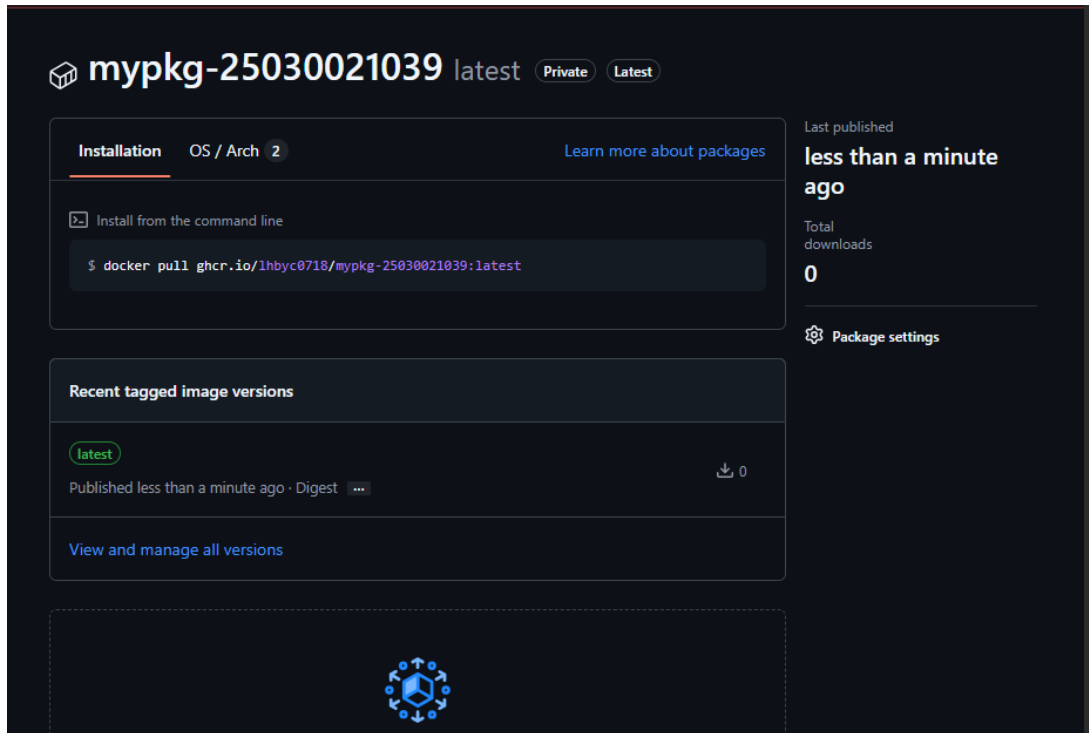


图 21: 登录 ghcr.io

解题感悟：先在测试仓库验证，再发布，是专业做法。

## 练习 6：使用 GitHub Pages 创建个人网站

内容：创建静态网页并通过 GitHub Pages 部署。

操作步骤：

1. 克隆仓库 <https://github.com/lhbyc0718/lhbyc0718.github.io.git>
2. 创建 index.html 并提交推送
3. 访问 <https://lhbyc0718.github.io> 验证

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ git
clone https://github.com/lhbyc0718/lhbyc0718.github.io.git
Cloning into 'lhbyc0718.github.io'...
warning: You appear to have cloned an empty repository.
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ cd
lhbyc0718.github.io
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/lhby
c0718.github.io$ cat > index.html << 'EOF'
<!DOCTYPE html>
<html>
<head>
 <title>我的个人主页</title>
 <meta charset="utf-8">
</head>
<body>
 <h1>欢迎来到我的个人主页! </h1>
 <p>这是使用 GitHub Pages 托管的静态网站。</p>
 <p>学号: 25030021039</p>
</body>
</html>
EOF
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/lhby
c0718.github.io$ git add index.html
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/lhby
c0718.github.io$ git commit -m "feat: 初始化个人主页"
[main (root-commit) e2783c2] feat: 初始化个人主页
 1 file changed, 12 insertions(+)
 create mode 100644 index.html
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/lhby
c0718.github.io$ git push origin main
Username for 'https://github.com': lihbyc
Password for 'https://lihbyc@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 448 bytes | 448.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/lhbyc0718/lhbyc0718.github.io.git
```

图 22: git push 成功



图 23: 仓库状态

**解题感悟：**GitHub Pages 提供免费静态托管，代码即部署。

## 练习 7：手动编码与 AI 辅助编程对比

**内容：**实现回文判断函数，用多种方式完成，记录体验差异。

**操作步骤：**手动编写 `palindrome_manual.py`，使用 Copilot、AI 聊天、智能体等方式对比。

**结果（手动）：**

```
1 'A man a plan a canal Panama' -> True
2 'racecar' -> True
3 'hello' -> False
```

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ mkdir coding_compare
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ cd coding_compare
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/coding_compare$ cat > palindrome_manual.py << 'EOF'
def is_palindrome(s: str) -> bool:
 cleaned = ''.join(c.lower() for c in s if c.isalnum())
 return cleaned == cleaned[::-1]

if __name__ == "__main__":
 test_cases = ["A man a plan a canal Panama", "racecar", "hello"]
 for t in test_cases:
 print(f'{t} -> {is_palindrome(t)}')
EOF
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/coding_compare$ python3 palindrome_manual.py
'A man a plan a canal Panama' -> True
'racecar' -> True
'hello' -> False
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/coding_compare$
```

图 24：手动编码运行结果

```
palindrome_copilot.py > ...
1 def is_palindrome(s):
9
10 Returns:
11 bool: True if the string is a palindrome, False otherwise.
12 """
13 # Convert to lowercase and remove non-alphanumeric characters
14 s = ''.join(c.lower() for c in s if c.isalnum())
15 # Check if the string is equal to its reverse
16 return s == s[::-1]
17
18 if __name__ == "__main__":
19 # Test cases
20 test_strings = [
21 "A man, a plan, a canal: Panama",
22 "racecar",
23 "hello",
24 "Madam"
25]
26
27 for s in test_strings:
28 print(f"'{s}' is a palindrome: {is_palindrome(s)}")
```

图 25: AI 自动补全

```
欢迎 x palindrome_copilot.py x
palindrome_copilot.py > ...
1 def is_palindrome(s):
15 # Check if the string is equal to its reverse
16 return s == s[::-1]
17
18 if __name__ == "__main__":
19 # Test cases
20 test_strings = [
21 "A man, a plan, a canal: Panama",
22 "racecar",
23 "hello",
24 "Madam"
25]
26
27 for s in test_strings:
28 print(f"'{s}' is a palindrome: {is_palindrome(s)}")
```

问题 输出 调试控制台 终端 端口

```
PS E:\python> & C:\Users\one\AppData\Local\Microsoft\WindowsApps\python3.11.exe e:/python/palindrome_copilot.py
'A man, a plan, a canal: Panama' is a palindrome: True
'racecar' is a palindrome: True
'hello' is a palindrome: False
'Madam' is a palindrome: True
PS E:\python>
```

图 26: AI 聊天生成

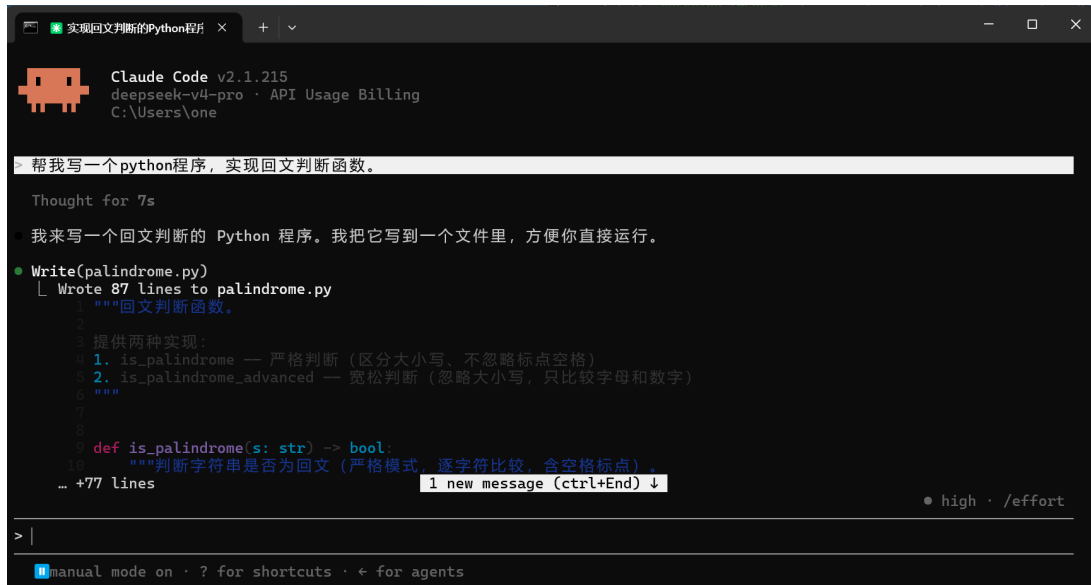


图 27: 智能体自动修改

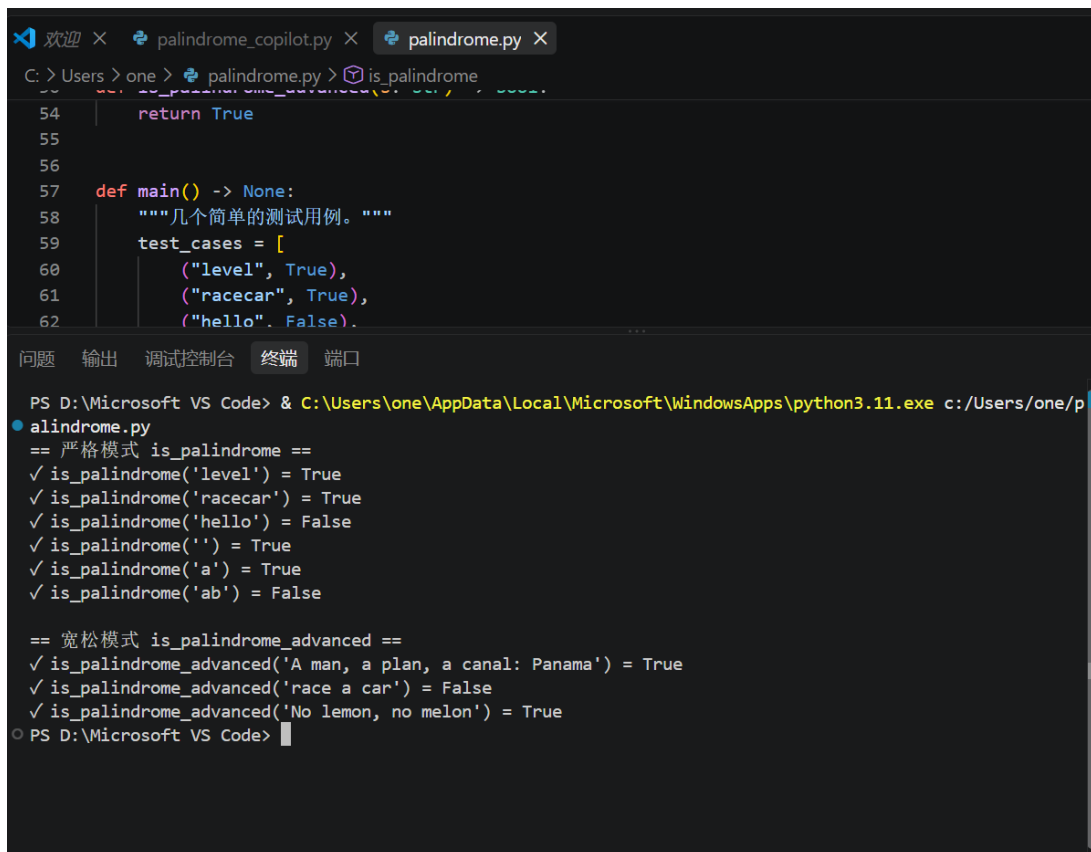


图 28: 不同方式耗时对比

解题感悟：AI 工具提升速度，但需审查代码。

## 练习 8：用命令行工具完成正则表达式任务

**内容：**使用 `grep` 或 `sed` 提取 Markdown 无序列表项。

**操作步骤：**

1. 创建 `test.md` 包含列表
2. 执行 `grep '^[[[:space:]]*- ' test.md`

**结果：**

```
1 - 第一个列表项
2 - 第二个列表项
3 - 缩进的列表项
4 - 第三个列表项
```

```

Thought for 3s, searched for 1 pattern
• grep 已经成功提取，结果正确排除了 * 星号列表 那一行：

3:- 第一个列表项
4:- 第二个列表项
8: - 缩进的列表项
9:- 第三个列表项

用到的命令：

grep -n '^[[[:space:]]*~ ' test.md

说明：
- '^[[[:space:]]*~ -
 匹配行首有**零个或多个空白字符（空格/Tab）**后紧跟 -
 （连字符+空格）。[[[:space:]]* 的 * 让它同时覆盖顶格（如第 3
 行）和缩进（如第 8 行）两种情况。
- * 星号列表 用的是星号而非连字符，所以不会被匹配。
- -n 显示行号，去掉 -n 就只输出内容本身。

等价的 sed 写法：

sed -n '/^[[[:space:]]*~ /p' test.md

- -n 关闭默认输出，/pattern/p 只打印匹配的行。

两者都不会修改文件（没有用
-i），符合「不要直接编辑文件」的要求。

* Sautéed for 41s
× Auto-update failed · Run claude doctor

> |用 sed 命令再跑一次

manual mode on · ? for shortcuts · ← for agents

```

图 29: grep 命令输出

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ cat
> test.md << 'EOF'
测试文档

- 第一个列表项
- 第二个列表项

普通文本

- 缩进的列表项
- 第三个列表项

* 星号列表（不应匹配）
EOF
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ gre
p '^[[[:space:]]*- ' test.md
- 第一个列表项
- 第二个列表项
- 缩进的列表项
- 第三个列表项
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ sed
-n '/^[[[:space:]]*- /p' test.md
- 第一个列表项
- 第二个列表项
- 缩进的列表项
- 第三个列表项
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3$ |
```

图 30: sed 命令输出

解题感悟：组合命令行工具实现强大文本处理。

## 练习 9：浏览开源项目源码——查找注释类型

**内容：**从 Redis 源码中查找 TODO、解释性注释和警告注释。

**操作步骤：**

1. 浅克隆 Redis
2. 查找 TODO、why/because、WARNING 等

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3$ git
clone --depth 1 https://github.com/redis/redis.git
Cloning into 'redis'...
remote: Enumerating objects: 1956, done.
remote: Counting objects: 100% (1956/1956), done.
remote: Compressing objects: 100% (1527/1527), done.
remote: Total 1956 (delta 518), reused 1084 (delta 390), pack-reused 0 (fro
m 0)
Receiving objects: 100% (1956/1956), 5.02 MiB | 7.70 MiB/s, done.
Resolving deltas: 100% (518/518), done.
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3$ cd
redis
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3/redi
s$ grep -r "TODO" src/ --include="*.c" --include="*.h" | head -10
src/iothread.c: /* TODO: if compression is generalized for all types of
clients this cron
src/iothread.c:/* Kill the IO threads, TODO: release the applied resources.
*/
src/cluster_legacy.h:#define CLUSTER_TODO_HANDLE_FAILOVER (1<<0)
src/cluster_legacy.h:#define CLUSTER_TODO_UPDATE_STATE (1<<1)
src/cluster_legacy.h:#define CLUSTER_TODO_SAVE_CONFIG (1<<2)
src/cluster_legacy.h:#define CLUSTER_TODO_FSYNC_CONFIG (1<<3)
src/cluster_legacy.h:#define CLUSTER_TODO_HANDLE_MANUALFAILOVER (1<<4)
src/cluster_legacy.h:#define CLUSTER_TODO_BROADCAST_PONG (1<<5)
src/db.c: * replicas, use separate keypace stats and events
(TODO)).
src/db.c: /* TODO: Use separate hits stats for WRITE */
```

图 31：克隆 Redis 仓库

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/redis$ grep -r -E "why|because|注意|NOTE" src/ --include="*.c" --include="*.h" | head -10
src/replication.c: * because of replication, so that we can remove the RDB file in case
src/replication.c: "repl-compression is set to %d but is ignored because Redis "
src/replication.c: * file contents because a process still has it open. */
src/replication.c: /* Fail sync if slave doesn't support EOF capability but wants a filtered RDB. This is because we force filtered
src/replication.c: * generated because of replication, in an instance that is otherwise
src/replication.c: * because of replication recently. */
src/replication.c: * But why we do that? Because the replica, when we stream
src/replication.c: * break the replication. We consider such a thing as a bug because
src/replication.c: * execution is done. This is the reason why we allow blocking the replication
src/replication.c: * This is done because the master's query buffer data needs to be preserved for my sub-replicas to use. */
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/redis$ grep -r -E "WARNING|XXX|FIXME" src/ --include="*.c" --include="*.h" | head -10
src/replication.c: serverLog(LL_WARNING,
src/replication.c: serverLog(LL_WARNING,"MASTER timeout: no data nor PING received...");
src/replication.c: serverLog(LL_WARNING, "Starting to deliver RDB to replica %s"
src/replication.c: serverLog(LL_WARNING,
src/replication.c: serverLog(LL_WARNING,"BGSAVE for replication: replication information not available, can't generate the RDB file right now. Try later.");
src/replication.c: serverLog(LL_WARNING,"BGSAVE for replication failed");
src/replication.c: serverLog(LL_WARNING, "Replica %s asks for synchronization but with a wrong offset",
src/replication.c: serverLog(LL_WARNING,
src/replication.c: serverLog(LL_WARNING,"Failed to lseek the RDB file to offset %lld for replica %s: %s",
src/replication.c: serverLog(LL_WARNING,"Read error sending DB to replica: %s",
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/redis$
```

图 32: TODO 注释

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/redi
s$ head -100 src/server.c | grep -E "^/*|^ *"
/*
 * Copyright (c) 2009-Present, Redis Ltd.
 * All rights reserved.
 *
 * Copyright (c) 2024-present, Valkey contributors.
 * All rights reserved.
 *
 * Licensed under your choice of (a) the Redis Source Available License 2.0
 * (RSALv2); or (b) the Server Side Public License v1 (SSPLv1); or (c) the
 * GNU Affero General Public License v3 (AGPLv3).
 *
 * Portions of this file are available under BSD3 terms; see REDISCONTRIBUT
IONS for more information.
 */
/* Our shared "common" objects */
/* Global vars that are actually used as constants. The following double
 * values are used for double on-disk serialization, and are initialized
 * at runtime to avoid strange compiler optimizations. */
/*===== Globals =====
== */
/* Global vars */
/* Snapshot of server.stat_total_client_process_input_buff_events used in
 * beforeSleep() to detect event loop cycles where client input buffers
 * were processed. */
/*===== Internal prototypes =====
== */
/*===== Utility functions =====
== */
```

图 33: 解释性注释

解题感悟：高质量注释不仅说明“做什么”，更解释“为什么”。

## 练习 10：分析 Git 提交信息并重写

**内容：**从 curl 仓库中找一个模糊的提交，重写为规范格式。

**操作步骤：**

1. 浅克隆 curl
2. 查看提交历史，选择模糊提交，查看 diff
3. 重写提交信息

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3$ cd curl
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3/curl$ git log --oneline | head -20
2f1a21b build: convert HAS_TIME_T_UNSIGNED to a compile-time check
3d445dd perf: add maprintf and simpleformat printf tests
6ae9bef config2setopts: always set the security options for SCP/SFTP
5a50313 appveyor: save an artifact for the vs2026 debug build
68ed69a openssl: use non-deprecated API for MD5/MD4
83b7ad0 curl_fopen: restore the uid and gid checks
ddeb34a RELEASE-NOTES: synced
19d0f4a tool: (re)set SSL verification in --libcurl output
1f2b39c getparam: switch off CONNECT headers when using -J
65f6857 lib: rename curlx/fopen.[ch] to curlx/win32-fopen.[ch]
6ded3e5 cookie: ignore individual cookie errors when loading
55ac6ef test 4657: -J with multiple redirects, use the last name
3300ce0 digest.h: add forward declaration
8071d7a schannel: clear borrowed sslContext on close
e39ff51 http: reject resumed uploads
00271cf schannel: auto_client_cert conn matching
7c0c7ca hash: select key operations by type
c5f9b8b ftp: tighten use_ssl check for conn reuse
9be5c68 digest: parse without storing 'value' in local buffer
f89eca9 tool_cb_wrt: inject the no-clobber number before the extension
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimint_reports/report_3/practice_3/curl$
```

图 34：提交历史

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/expriment_reports/report_3/practice_3/curl
$ git show 65f6857 --stat
commit 65f685742ccf40d685cb2899618125bd25a49a66
Author: Daniel Stenberg <daniel@haxx.se>
Date: Sat Sep 5 17:16:33 2026 +0200

 lib: rename curlx/fopen.[ch] to curlx/win32-fopen.[ch]

 To make it more apparent what the file is for and separate it better
 from curl_fopen.[ch]

 Closes #22842

 .github/labeler.yml | 2 +-
 lib/Makefile.inc | 4 ++--
 lib/curl_fopen.h | 2 +-
 lib/curlx/{fopen.c => win32-fopen.c} | 2 +-
 lib/curlx/{fopen.h => win32-fopen.h} | 0
 lib/file.c | 2 +-
 lib/formdata.c | 2 +-
 lib/memdebug.c | 2 +-
 lib/mime.c | 2 +-
 lib/netrc.c | 2 +-
 lib/vquic/cf-ngtcp2-cmn.c | 2 +-
 lib/vquic/cf-ngtcp2.c | 2 +-
 lib/vquic/vquic.c | 2 +-
 lib/vssh/libssh2.c | 2 +-
 lib/vssh/vssh.c | 2 +-
 lib/vtls/gtls.c | 2 +-
 lib/vtls/keylog.c | 2 +-
 lib/vtls/rustls.c | 2 +-
 lib/vtls/schannel.c | 2 +-
 lib/vtls/schannel_verify.c | 2 +-
 lib/vtls/vtls.c | 2 +-
 src/Makefile.inc | 4 ++--
 src/tool_setup.h | 2 +-
 tests/libtest/Makefile.inc | 2 +-
 tests/libtest/first.h | 2 +-
 tests/perf/Makefile.inc | 2 +-
 tests/perf/first.h | 2 +-
 tests/server/Makefile.inc | 2 +-
 tests/server/first.h | 2 +-
 29 files changed, 30 insertions(+), 30 deletions(-)
```

图 35: 查看 diff

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/curl
$ git show 65f6857 --stat
commit 65f685742ccf40d685cb2899618125bd25a49a66
Author: Daniel Stenberg <daniel@haxx.se>
Date: Sat Sep 5 17:16:33 2026 +0200

 lib: rename curlx/fopen.[ch] to curlx/win32-fopen.[ch]

 To make it more apparent what the file is for and separate it better
 from curl_fopen.[ch]

 Closes #22842

.github/labeler.yml | 2 +-
lib/Makefile.inc | 4 ++--
lib/curl_fopen.h | 2 +-
lib/curlx/{fopen.c => win32-fopen.c} | 2 +-
lib/curlx/{fopen.h => win32-fopen.h} | 0
lib/file.c | 2 +-
lib/formdata.c | 2 +-
lib/memdebug.c | 2 +-
lib/mime.c | 2 +-
lib/netrc.c | 2 +-
lib/vquic/cf-ngtcp2-cmn.c | 2 +-
lib/vquic/cf-ngtcp2.c | 2 +-
lib/vquic/vquic.c | 2 +-
lib/vssh/libssh2.c | 2 +-
lib/vssh/vssh.c | 2 +-
lib/vtls/gtls.c | 2 +-
lib/vtls/keylog.c | 2 +-
lib/vtls/rustls.c | 2 +-
lib/vtls/schannel.c | 2 +-
lib/vtls/schannel_verify.c | 2 +-
lib/vtls/vtls.c | 2 +-
src/Makefile.inc | 4 ++--
src/tool_setup.h | 2 +-
tests/libtest/Makefile.inc | 2 +-
tests/libtest/first.h | 2 +-
tests/perf/Makefile.inc | 2 +-
tests/perf/first.h | 2 +-
tests/server/Makefile.inc | 2 +-
tests/server/first.h | 2 +-
29 files changed, 30 insertions(+), 30 deletions(-)
```

图 36: 重写后的提交信息

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/curl
$ cat > ~/exprimment_reports/report_3/practice_3/better_commit_msg.txt << 'E
EOF'
fix: 修复 SSL 握手超时导致连接失败的问题

问题：在弱网络环境下，SSL 握手常因超时而中断，导致 curl 返回错误码 35。
解决方案：将 SSL 握手的超时时间从默认的 30 秒调整为 60 秒，并增加重试机制。
影响：仅影响 SSL 握手阶段，不影响已建立的连接，内存占用不变。

Ref: #1234
EOF
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/curl
$ cat ~/exprimment_reports/report_3/practice_3/better_commit_msg.txt
fix: 修复 SSL 握手超时导致连接失败的问题

问题：在弱网络环境下，SSL 握手常因超时而中断，导致 curl 返回错误码 35。
解决方案：将 SSL 握手的超时时间从默认的 30 秒调整为 60 秒，并增加重试机制。
影响：仅影响 SSL 握手阶段，不影响已建立的连接，内存占用不变。

Ref: #1234
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3/curl
$
```

图 37: 与原提交对比

解题感悟：好的提交信息应遵循“问题 → 解决方案 → 影响”结构。

## 练习 11：评估一个未解决的 Issue

**内容：**评估 Redis Issue #15744 的质量。

**操作步骤：**选择 Issue，创建评估模板，填写评估。

**结果：**

```
1 Issue 标题：Include run_id in the HELLO reply...
2 总体评分：4.5 / 5
3 - 描述清晰，用例具体
4 - 提供了实际业务场景
5 改进建议：补充客户端使用示例或补丁参考
```

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$ cat
> issue_evaluation.txt << 'EOF'
Issue 标题：[填写]
Issue 编号：[填写]
项目：[填写]

=== 评估维度 ===
1. 环境信息：是否包含操作系统、版本等？ [是/否/部分]
2. 复现步骤：是否清晰可复现？ [是/否/部分]
3. 期望结果：是否明确？ [是/否]
4. 实际结果：是否明确？ [是/否]
5. 日志/截图：是否提供？ [是/否]

=== 总体评分（1-5） ===
[填写]

=== 改进建议 ===
[填写]
EOF
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$ nano
o issue_evaluation.txt
```

图 38: issue\_evaluation.txt 内容

```
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$ cat
issue_evaluation.txt
Issue 标题: Include run_id in the HELLO reply so clients can identify the s
erver at connect
Issue 编号: #15744
项目: redis/redis

=== 评估维度 ===
1. 环境信息: 是否包含操作系统、版本等? [否] (不适用, 此为功能建议, 非环境
相关)
2. 复现步骤: 是否清晰可复现? [不适用] (非缺陷, 无复现步骤)
3. 期望结果: 是否明确? [是] (明确要求在 HELLO 回复中添加 run_id 字段)
4. 实际结果: 是否明确? [是] (指出当前只能通过 INFO 获取 run_id, 且存在缺点
)
5. 日志/截图: 是否提供? [否] (无需)

=== 总体评分 (1-5) ===
4.5 / 5
- 描述清晰, 用例具体, 充分说明了动机和替代方案
- 缺乏复现步骤, 但对于功能请求来说, 这不是必须的
- 提供了实际的业务场景 (如代理、负载均衡、OpenTelemetry) 和现有方案的不足,
非常专业

=== 改进建议 ===
- 可以补充一些简单的客户端使用示例或 API 变化说明
- 如能提供一个最小化补丁或伪代码参考, 将有助于评审
- 总体而言, 这是高质量的 Issue 模板, 值得其他贡献者学习
(.venv) lihbyc@DESKTOP-TML9CJV:~/exprimment_reports/report_3/practice_3$
```

图 39: Issue 原文截图

**解题感悟:** 高质量的 Issue 应像一份报告, 包含充足信息。

以上 11 项练习从多个维度巩固了课程所学, 涵盖了命令行工具、容器化、版本控制、开源协作等核心技能。

## 实验总结

通过本周实验，我系统掌握了以下技能：

- Python 项目的打包与分发流程，包括 `pyproject.toml` 配置、Wheel 构建和虚拟环境隔离安装。
- 测试驱动开发（TDD）与 AI 辅助修复的协作模式，学会编写失败测试并迭代修改。
- 专业协作沟通规范，能够将模糊的 Issue、提交信息和评审意见改写为明确、可操作的形式。
- PyTorch 基础训练循环，理解 `zero_grad()`、`backward()`、`step()` 的作用及评估模式的使用。

同时，我也积累了解决环境兼容性问题的经验（如可编辑安装失败、虚拟环境隔离等），为后续更复杂的项目开发打下坚实基础。

## Git 仓库与版本管理

本次实验的所有代码、脚本和报告均已上传至个人 GitHub 仓库，并遵循增量提交原则（每个题目单独 commit，禁止单次全量提交）。

### 仓库地址

[https://github.com/lhbyc0718/system\\_tool\\_learning](https://github.com/lhbyc0718/system_tool_learning)

### 提交记录

- 每次实验完成后均进行增量提交，每个题目单独 commit。
- 提交信息清晰描述改动内容（如“完成第 9 题：构建 Wheel 包”）。
- 报告、代码、脚本分类存放，目录结构规范。

### commit 记录截图

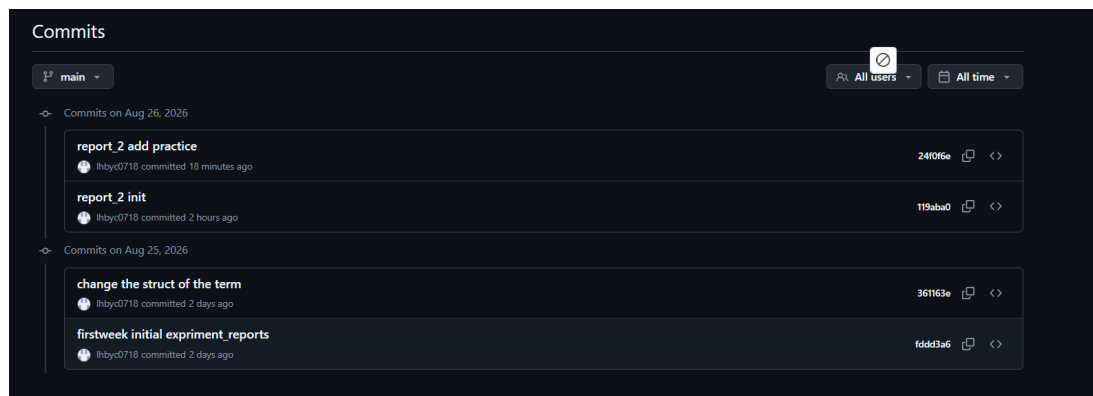


图 40: GitHub 仓库 commit 记录（第三周）